Description

Illustration



Figure caption

Bitumen is a viscous black material that sets on cooling to form a waterproof coating. 1. Liquid bitumen being poured into a tin. 2. Bitumen being applied to a roof as waterproofing. Images used under license from Shutterstock.com

The material

Bitumen (asphalt, tar) is a highly viscous, black, sticky organic material, wholly soluble in carbon disulfide. It occurs naturally in bitumen lakes (Trinidad) but most is now derived as a by-product of oil refining. Its mechanical properties are characterized by the "Pen" value with units of mm, standing for "Penetration depth", an indentation test used to characterize mechanical response. The mechanical properties vary strongly with temperature, time of loading and with the grade of bitumen. The data below are for 20 C, the moduli are for dynamic loading and the strengths are for a loading time of about 1000 seconds. All must be regarded as approximate.
Bitumen is used as a binder in asphalt concrete for road beds, as a sealing agent for water exclusion, and as the sealing agent in tar felts and tar paper.

Compositional summary

Complex hydrocarbons:

General properties

Density	997	-	1.05e3	kg/m^3
Price	0.226	-	0.34	GBP/kg
Date first used	-4000			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓

Interiors	✖
Services	✖

Mechanical properties

Young's modulus	* 1.5	-	2.7	GPa
Shear modulus	* 0.5	-	0.9	GPa
Bulk modulus	* 1.8	-	3	GPa
Bending modulus	* 1.5	-	2.7	GPa
Poisson's ratio	* 0.4	-	0.42	
Yield strength (elastic limit)	* 1	-	1.4	MPa
Tensile strength	* 1	-	1.4	MPa
Compressive strength	* 1	-	1.4	MPa
Bending strength	* 1	-	1.4	MPa
Elongation	* 10	-	30	% strain
Hardness - Vickers	* 0.3	-	0.4	HV
Fatigue strength at 10^7 cycles	* 0.3	-	1	MPa
Mechanical loss coefficient (tan delta)	* 0.8	-	1	

Thermal and combustion properties

Thermal conductor or insulator?	Poor insulator			
Thermal resistivity	0.17	-	0.2	m.°C/W
Thermal expansion coefficient	600	-	620	µstrain/°C
Specific heat capacity	1.68e3	-	1.8e3	J/kg.°C
Maximum service temperature	64	-	88	°C
Flammability	Highly flammable			
Emissivity	0.93	-	0.95	

Hygro-thermal properties

Water absorption	0			%
Water vapor permeability	0			kg.m/s.m^2.Pa
Frost resistance	Good			

Electrical properties

Electrical conductor or insulator?	Good insulator			
Electrical resistivity	1e12	-	1e14	µohm.cm
Dielectric constant (relative permittivity)	2.7	-	3	
Dissipation factor (dielectric loss tangent)	0.005	-	0.05	
Dielectric strength (dielectric breakdown)	10	-	20	MV/m

Optical properties

Natural color

Black

Transparency	Opaque	
Transmissivity	0	%

Critical Materials Risk

High critical material risk?	No

Acoustic properties

Sound absorption	Good
Sound isolation	Average

Processability (scale 1 = impractical to 5 = excellent)

Castability	5
Moldability	5

Durability

Water (fresh)	Excellent
Water (salt)	Excellent
Weak acids	Excellent
Strong acids	Limited use
Weak alkalis	Excellent
Strong alkalis	Acceptable
Organic solvents	Unacceptable
UV radiation (sunlight)	Fair
Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	*	0.2	-	0.3	kg/kg
Embodied energy (virgin grade production)		101	-	111	MJ/kg

Recycling and end of life

Recycle	✗				
Recycle fraction in current supply		0.1			%
Combust for energy recovery	✓				
Heat of combustion (net)	*	41	-	48.5	MJ/kg

Combustion CO2	* 2.46	-	2.91	kg/kg
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Bitumen, and the fumes it emits when heated, are toxic.

Technical notes

Bitumen is composed of asphaltenes; polar condensed aromatics (5-25%); polar aromatic maltenes (15-30%); non-polar aromatic maltenes (40-65%); aliphatic hydrocarbon maltenes and alkyl substituted cyclic maltenes (5-20%).

Typical uses

Damp proofing of floors, tiles and singles, tar felts and tar paper for water exclusion and insulation, treatment of timber, road beds and pavements.

Links

ProcessUniverse

Reference